Two Samples of Service Times

Unit 4 · Topic 4.7 · TODAY'S PROBLEM

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE (5 min, on your sheet)

Does sorting the times make the North and South records real pairs? Explain in one sentence.

DOK 1 **(a)** Find the North-minus-South difference in sample mean service times.

DOK 2 **(b)** The independent-sample standard error is 1.88797 minutes. Use $t^* = 1.995$ to calculate the 95% interval.

DOK 3 **(c)** In 3–4 sentences, choose a procedure using the original sampling design. Compare the two intervals and explain how sorting the times could mislead a reader about whether the population means differ.

Video + follow-along: `u7_lesson6_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.